



MBBS IV Junior Clerkship

Lectures 44

Splenomegaly:

Common causes of splenomegaly
Myeloproliferative Neoplasm

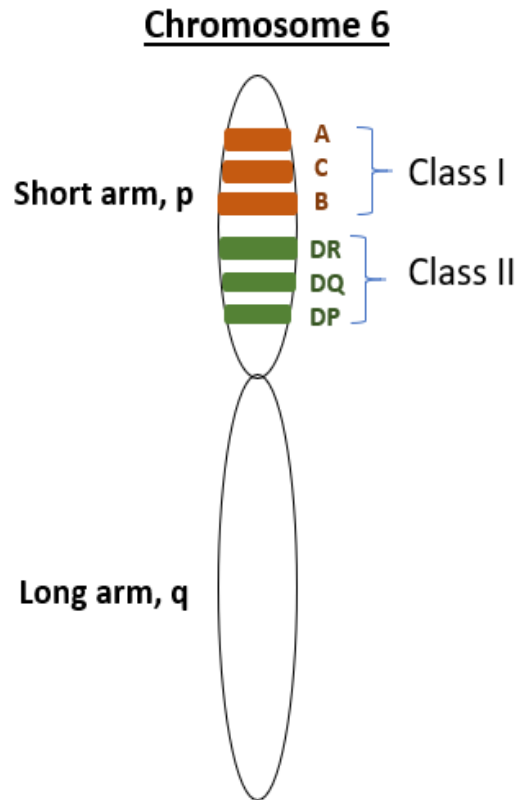
Joycelyn Sim

Department of Medicine

Queen Mary Hospital

Dec 2023

Human Leukocyte Antigen (HLA) System



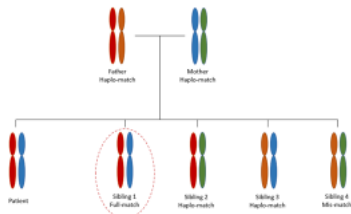
- Encoded by the **Major Histocompatibility Complex (MHC)** gene on the short arm of chromosome 6
- The HLA antigens are expressed on cell surface and involved in recognition of **“self” (tolerate)** and **“non-self” (eliminate)**.
 - **Class I** present on all nucleated cells
 - **Class II** present on immune cells
- HLA matching is crucial in HSCT to avoid graft rejection and graft versus host disease

Take home message

High white cell count

- **Reactive** causes are common in clinical practice
- Ascertainment of the abnormal **cell type**
- Leukaemia treatment depends on specific **diagnosis, disease risk and patient comorbidities**
- Look out for **haematological emergencies**
- **Allogeneic HSCT** for high risk cases and at relapse
- **HLA-matching** is essential in allogeneic HSCT

HLA compatibility within family



- The HLA genes are inherited from parents in a **simple Mendelian co-dominant** fashion
- Each sibling inherited **one haplotype from each parent**
- Each sibling has a **1 in 4 chance** to be HLA identical to the patient
- **HLA-identical by descent**

Who can be donor for HSCT?

Depending on donor type

	Donor
Autologous	Self
Syngeneic	Monoczygotic twins
Allogeneic	HLA matched sibling
	HLA matched unrelated donor
	HLA matched cord blood
	Haploidentical (mismatched family) donor

Objective

- Understand the **pathophysiology** of splenomegaly
- Understand the **common causes** of splenomegaly
- Understand the **clinical** and **laboratory** features of splenomegaly
- Acquire the ability to discuss the **treatment options** of splenomegaly
- Myeloproliferative Neoplasm

Two years ago....

Lecture 11: Structure and function of lymphatic organs

- Identify surface anatomy and describe the topography of lymphatic system.
- Explain the architecture and function of lymph nodes.
- List the structures and functions of other lymphoid tissues (tonsils, Peyer's patches).
- Describe the thymic structure and histological organization.
- Identify the topography, and describe the structure and function of the spleen.

[Dr CYH See, Biomedical Sciences]

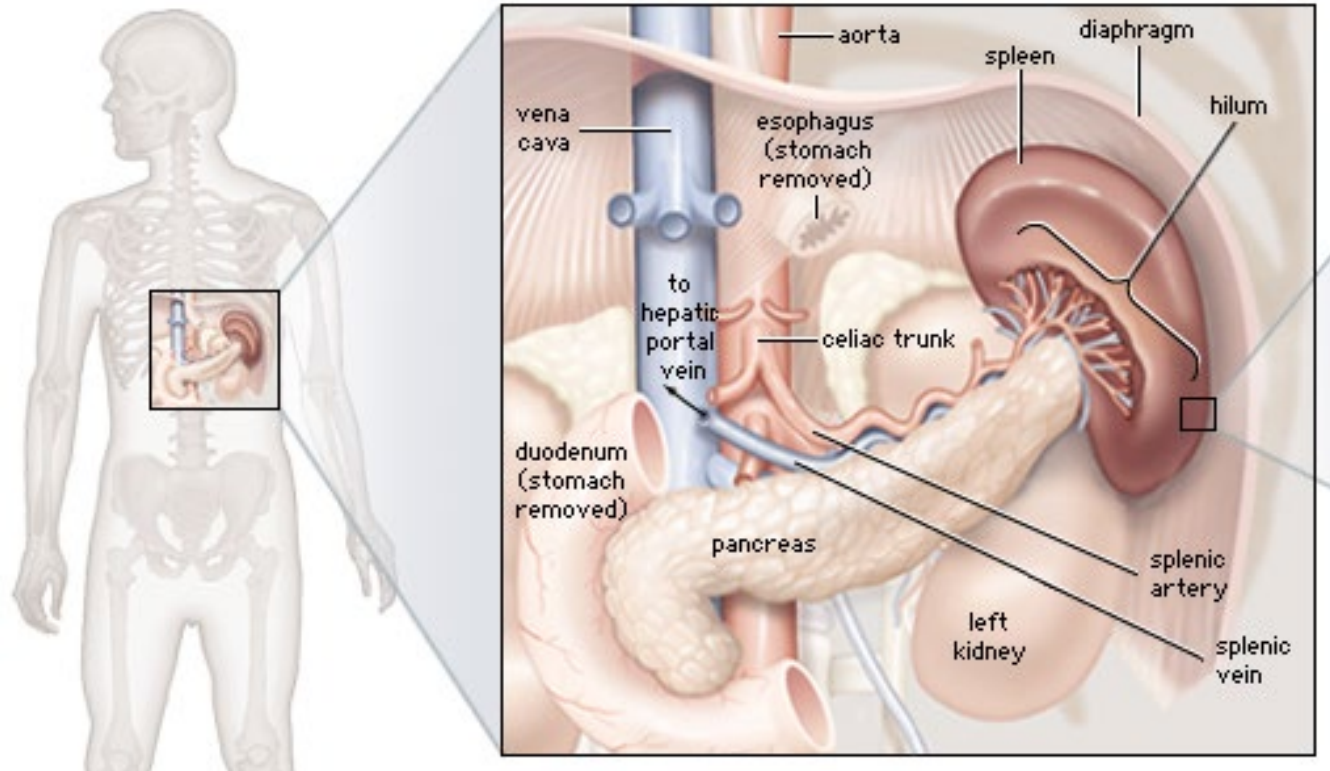
Lecture 24: Myelodysplastic syndromes and myeloproliferative diseases: principles of management

- Demonstrate understanding of the classification of the myelodysplastic syndrome and myeloproliferative neoplasms.
- Demonstrate understanding of the principles of targeted therapy.
- Demonstrate understanding of the principles of supportive treatment.

[Prof YL Kwong, Medicine]

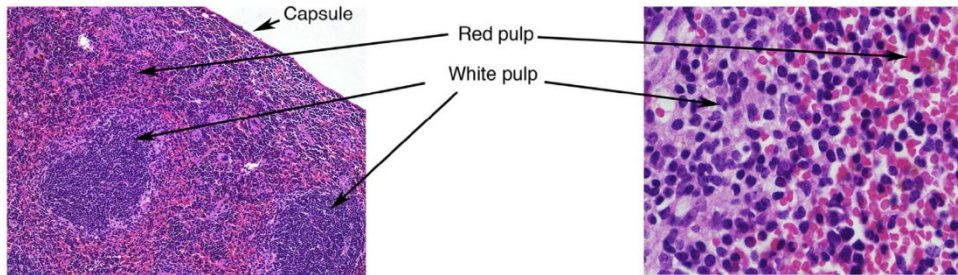
Anatomy of normal spleen

- Lies within the peritoneal cavity in the LUQ
- Adjacent to ribs **9,10,11**
- Size correlates with **person's height and weight**



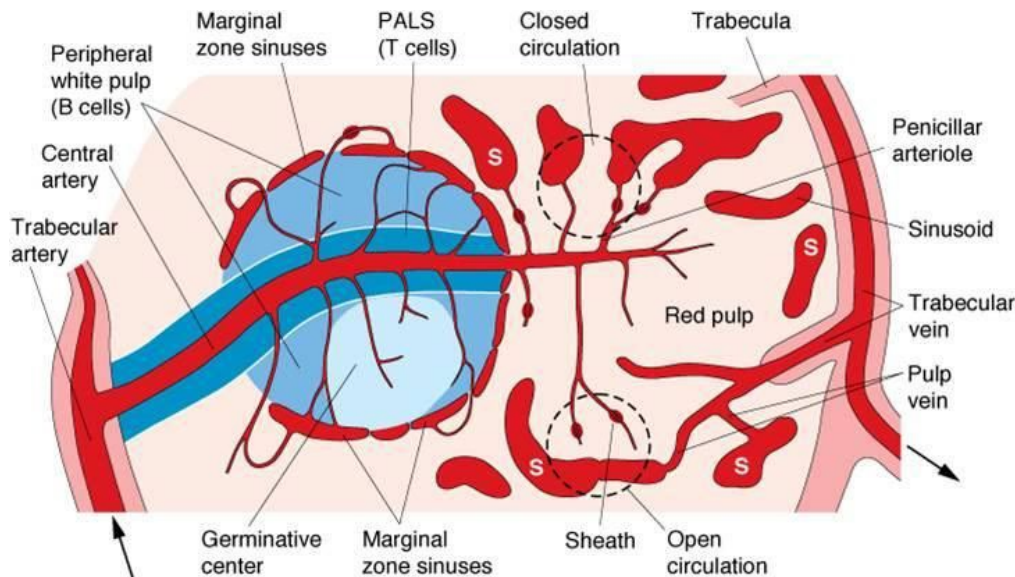
A local study recommended 12 cm as the upper limit of normal in HK Chinese Loftus et al. J Ultrasound Med 1997.

Anatomy of normal spleen



1. Red pulp

- Microcirculatory structure of **venous sinuses and splenic cords** (“Cords of Billroth”)



2. White pulp

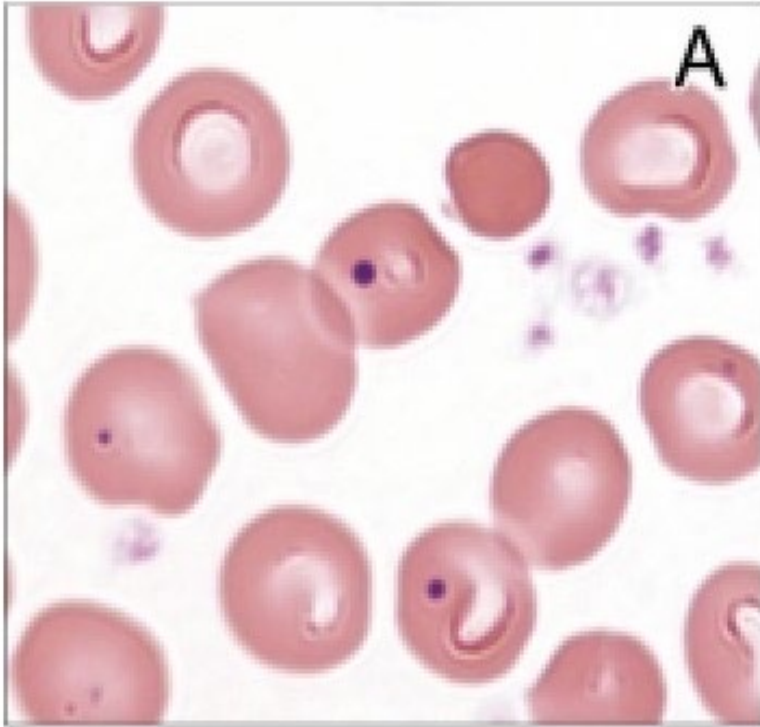
- peri-arteriolar **lymphoid sheath (PALS)** contains mainly **T-cells (CD4)** and the **primary follicles** contains B-cells



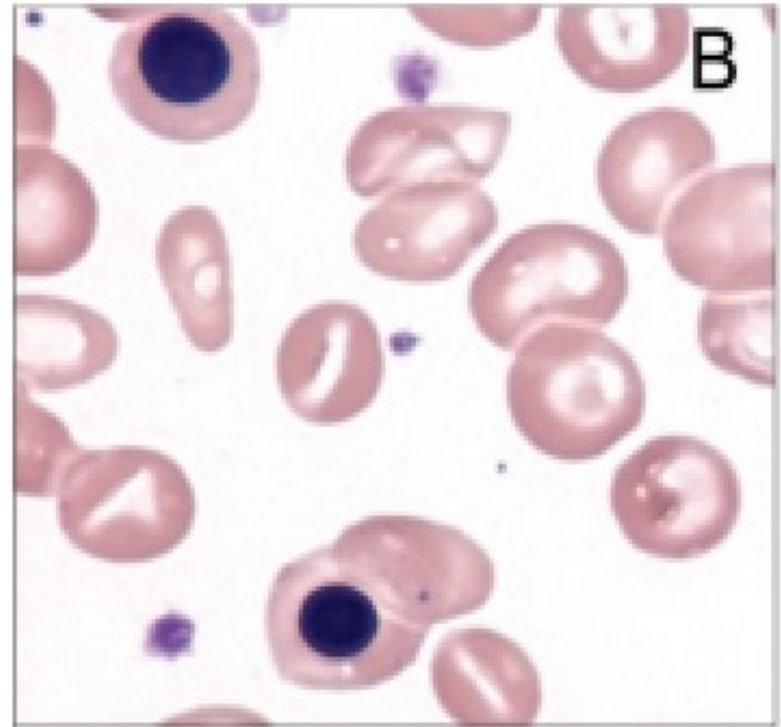
Normal function of spleen

1. **Filtration** of circulating blood
 - Old senescent RBC (culling)
 - Remove abnormal RBC inclusions e.g. Howell Jolly bodies
2. Site of **immune response**
 - Antibodies production
 - Blood-borne pathogens
3. **Storage of blood**
 - 5% red cell mass, 30-50% WBC margining pool, 20-40% total platelet mass
4. **Haematopoiesis (only in the fetus)**

Post splenectomy/hyposplenism



Howell Jolly bodies



Nucleated Red blood cells

Post splenectomy/hyposplenism



How to examine for splenomegaly

1. LUQ mass
2. Moves down with respiration along **Gardner's line**
3. Cannot get above it
4. Dull to percussion
5. Splenic notch



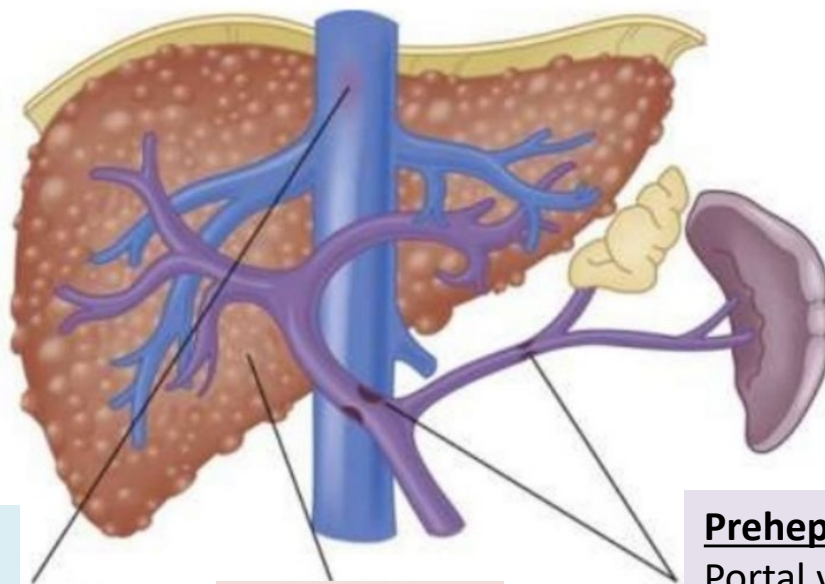
Mechanism of splenomegaly [1]

INCREASED SPLENIC FUNCTION	Pathophysiology	Examples
Increased “red pulp” function	Destruction of abnormal RBC from circulation	Haemolytic anaemia E.G Hereditary spherocytosis Thalassaemia, HbH disease AIHA
Increased “white pulp” function	Infection	Malaria Tuberculosis EBV, CMV, HIV
	Immune inflammation	Rheumatoid arthritis with Felty syndrome

Mechanism of splenomegaly [2]

VASCULAR CONGESTION

Blockage of blood flow and pooling of blood in red pulp



Posthepatic

Budd Chiari syndrome
IVC obstruction
Right heart failure

Intrahepatic

Liver cirrhosis

Prehepatic

Portal vein thrombosis
Splenic vein thrombosis

Mechanism of splenomegaly [3]

INFILTRATION	Examples
Haematological	Myeloproliferative neoplasm (MPN) E.g. Primary myelofibrosis Polycythaemia vera Essential thrombocythaemia Lymphoma
Non-haematological	Glycogen storage disease E.g. Gaucher's

Sizes

of splenomegaly are related to the causes of splenomegaly

Size of spleen	Causes of splenomegaly
Massive splenomegaly	Chronic myeloid leukaemia Myelofibrosis
Moderate splenomegaly	Portal hypertension Haematological malignancies
Minimal splenomegaly	Haemolytic anaemia • Thalassaemia intermedia • HbH disease • Autoimmune cytopenia (ITP, AIHA)

Geographical

location determines the possible causes of splenomegaly

Demark	Pakistan Tertiary Care
Haematological 39%	Chronic liver disease 64%
Liver disease 18%	Malaria 16%
Infection 10%	Haematological malignancies 14%

Rotbain et al. PLOS ONE 2017

Ghalaut et al. Intern J Sci Res pub 2016

Humaira et all. J Liaquat Uni Med Heath Sci 2016

Disease Prevalence

Causes of splenomegaly		Common causes of splenomegaly
Neoplasia	Lymphomas, Leukaemias, Myeloproliferative disorders	
Portal hypertension	Cirrhosis Splenic venous obstruction	
Haemolytic anaemia	Thalassaemia intermedia / major Autoimmune haemolytic anaemia Hereditary spherocytosis	

Causes of splenomegaly		Other rare causes of splenomegaly
Chronic inflammatory conditions	Autoimmune diseases	
Infections	Infective endocarditis Schistosomiasis Malaria	

Clinical features of splenomegaly

Pressure symptoms from the enlarged spleen

1. Abdominal fullness
2. Early satiety
3. Symptoms related to pancytopenia

Features due to Underlying Diseases

Features of cirrhosis ➤ stigmata of chronic liver disease

Features of portal hypertension ➤ ascites, caput medusae, esophageal varices

Laboratory features of splenomegaly

Abnormal blood counts ➤ anaemia
➤ thrombocytopenia
➤ leucopenia

Abnormal liver function tests ➤ cirrhosis

What is hypersplenism?

Hypersplenism: ↑ blood sequestration

1. Pancytopenia
2. Splenomegaly
3. Evidence of an **active marrow**
 - ❖ Patients with an abnormal marrow e.g. due to primary myelofibrosis, may develop pancytopenia due to splenomegaly
 - ❖ This should not be called hypersplenism
4. Reversal of abnormal features on splenectomy (if performed)

Pancytopenia + splenomegaly ≠ hypersplenism

Approach to a patient with splenomegaly

Complete history

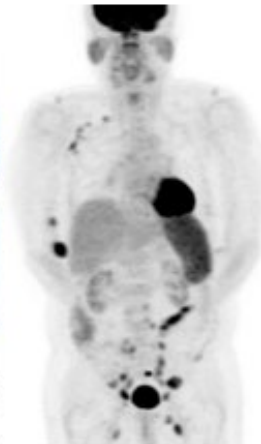
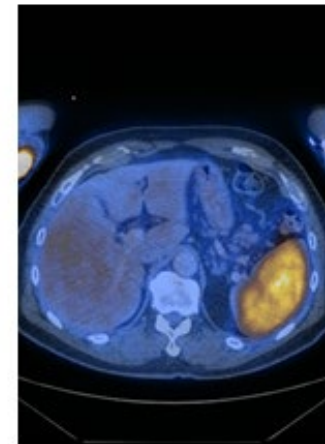
1. Age, gender, race
2. Residence & travel history
3. Alcohol
4. Constitutional symptoms: fever, weight loss, sweating
5. Bleeding tendency

Complete physical examination

1. Size of the spleen
2. Associated hepatomegaly, lymphadenopathy
3. Stigmata of chronic liver disease
4. Jaundice
5. Fever or other sign of infection
6. Cardiac murmur, splinter hemorrhage

Investigations

1. CBP, LRFT, virological studies
2. Imaging e.g. USG, CT, PET-CT
 - Confirm spleen size
 - Splenic architecture (focal lesions vs diffuse splenomegaly)
 - Assess other organs (e.g. liver cirrhosis)



Case 1

- M/60, Heavy drinker
- Mother died of HBV related HCC
- Presented with jaundice, tea colour urine, and abdominal distension for 2 months
- Abdominal examination showed dilated abdominal veins, splenomegaly 4cm below left costal margin, liver not palpable, shifting dullness
- **What is the cause for the splenomegaly?**

Case 2

- M/20, Medical student, Good past health
- Presented with fever, and cervical lymphadenopathy for 2 week
- Abdominal examination showed hepatomegaly 2cm, and splenomegaly 2cm
- Monospot test was positive
- EBV VCA IgM was positive
- **What is the cause of the splenomegaly?**

Case 3

- F/30. Good past health. Travelled to Philippines islands 3 weeks ago for diving
- Did not take anti-malarial prophylaxis
- Come back with fever, dark colour urine
- Physical examination showed pallor, jaundice, splenomegaly 3cm below costal margin
- Blood film showed plasmodium falciparum
- **What is the cause of the splenomegaly?**

Case 4

- F/60, Good past health
- Presented with weight loss and loss of appetite. No fever
- Physical examination showed pallor and massive splenomegaly up to the umbilicus
- CBP showed pancytopenia with leukoerythroblastic blood picture and tear drop cells
- **What is the cause of the splenomegaly?**

Therapy

Therapeutic options depends on the underlying cause

Indication for splenectomy

1. Diagnostic

- Suspected splenic lymphoma

2. Therapeutic

- Symptomatic relief in massive splenomegaly e.g. myelofibrosis (less common)
- Refractory ITP or AIHA (less common)
- Decrease transfusion requirement in thalassaemia major

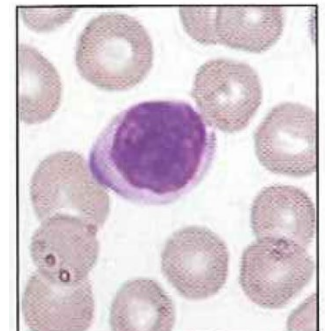
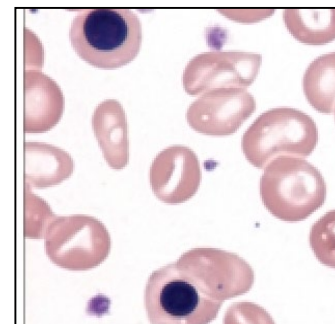
Issues in Splenectomy

Risk of **INFECTION**

Life-threatening infection from Encapsulated bacteria	
Pre - operative vaccination: 1. pneumococcus 2. H. influenza 3. meningococcus	Post - operative Antibiotic prophylaxis
Patient education Immediate referral to a hospital for treatment	

Increased cell counts

- Spurious leucocytosis after splenectomy
- Increase in platelet counts

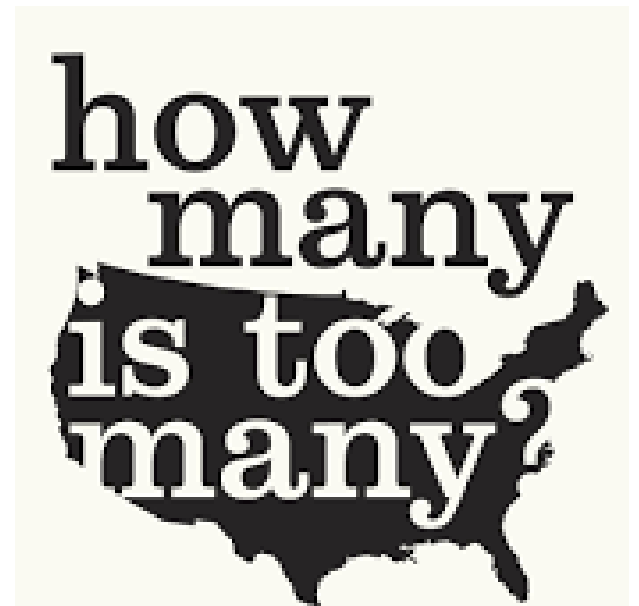


Vaccination for splenectomy patients

- **Overwhelming post splenectomy infection syndrome (OPSI)**
 - a **rare** but **life-threatening** complication in post-splenectomy patients
 - caused by invasive infections of **encapsulated bacteria**, >50% of the infections are due to **Streptococcus pneumoniae**.
 - life-time risk of OPSI is estimated to be **5%** but most (50%) occur within the **first two years of splenectomy**
 - prevention of OPSI include **vaccination** and **early empirical antibiotics** for febrile episodes
- Vaccinations are recommended against **S. pneumoniae**, **N. meningitidis**, **H. influenzae** type b and **influenza** virus
- For elective splenectomy, the first vaccine should be administered **at least 2 weeks before surgery**
- For emergency splenectomy, vaccines can be administered **2 weeks after the surgery**



Myeloproliferative Neoplasm



Epidemiology

Estimated Incidence in Hong Kong

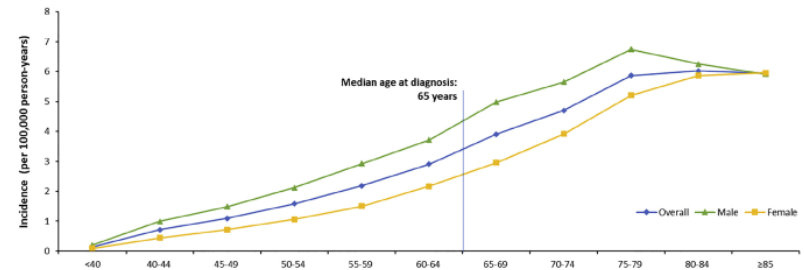


7,500,000

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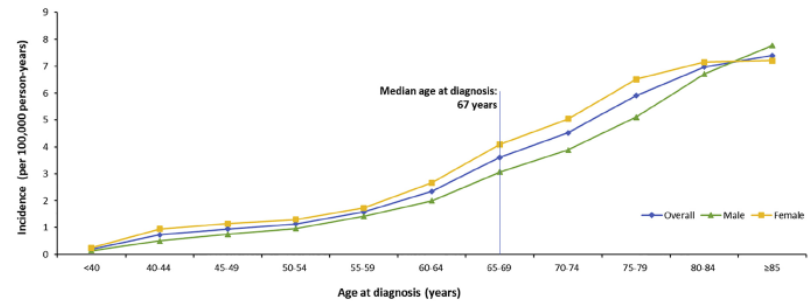
Epidemiology

Polycythemia Vera



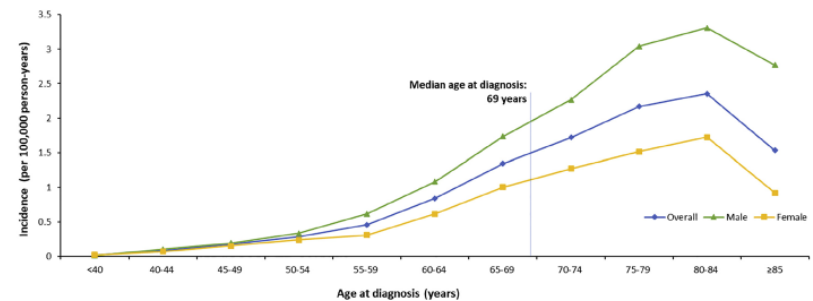
Epidemiology

Essential Thrombocythemia



Epidemiology

Primary Myelofibrosis



Myeloproliferative neoplasms (MPN)

MPN are **clonal** haematopoietic stem cell disorders characterized by the **overproduction** of **mature** cells of myeloid lineages.

Phenotype	Molecular marker	
Chronic myeloid leukaemia	BCR-ABL1	BCR-ABL neg
Polycythaemia vera	JAK2 mutations (>95%)	
Essential thrombocythaemia	JAK2 mutation (50%)	
Primary myelofibrosis	JAK2 mutations (50%)	

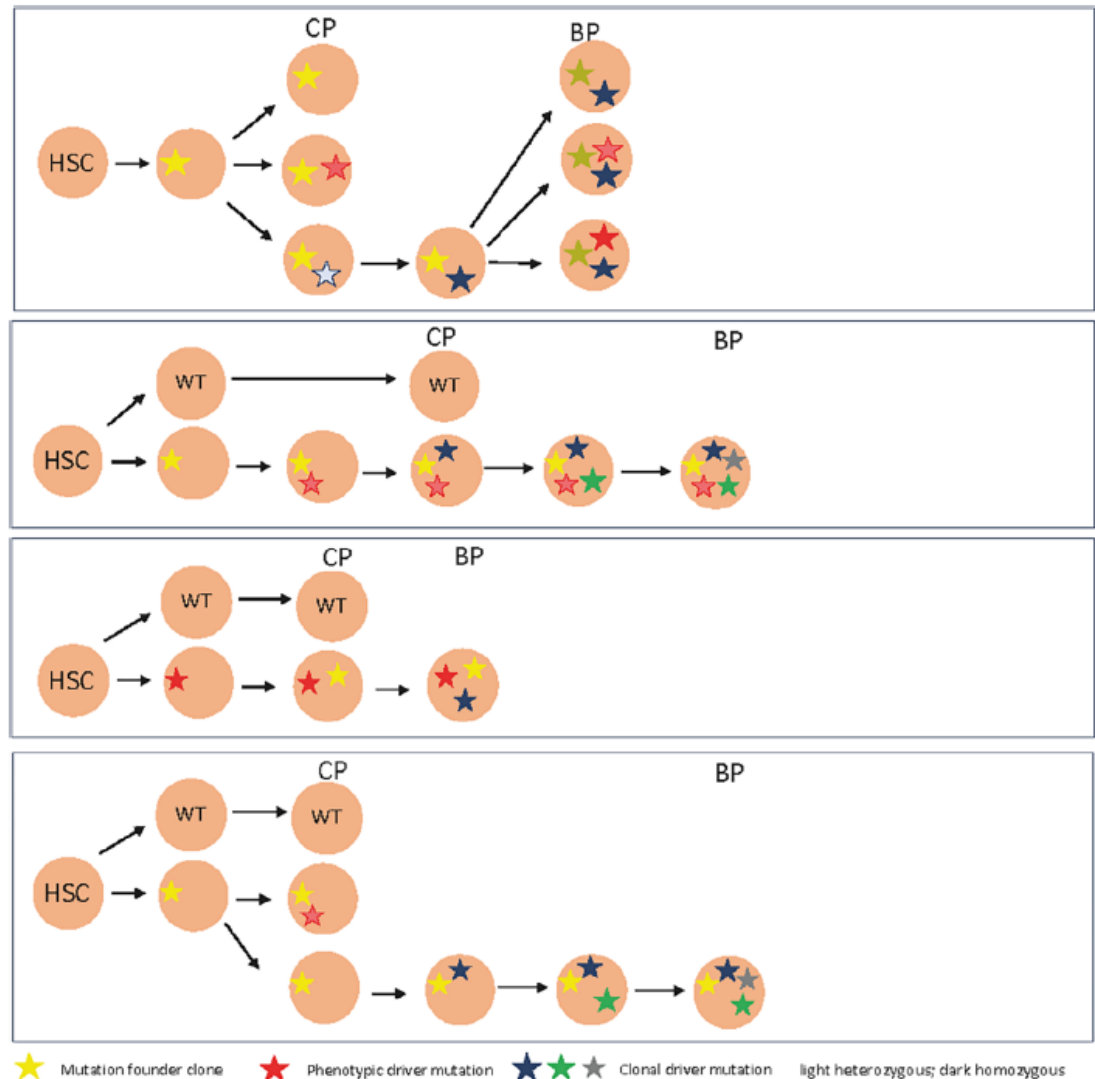
Share potential to **PROGRESS** to

1. Myelofibrosis (PPVMF, PETMF) and
2. Blastic transformation

Molecular pathogenesis of myeloproliferative neoplasms: Where do we stand in 2023?

Violaine Havelange^{1,2} | Stefan N. Constantinescu^{1,2,3,4,5} 

Patterns of clonal evolution
from the HSC to the chronic
phase (CP) MPN and blast
phase (BP)





WHO and IWG-MRT Criteria 2022

PV	Post-PV MF	ET	Post-ET MF
Major Criteria			
• ↑ Hb, Hct • BM morphology • JAK2 V617F or exon 12	• Prior PV • BM fibrosis 2-3	• Platelet $\geq 450 \times 10^9/L$ • BM morphology • Not meeting criteria for other MDS/MPN • JAK2, CALR or MPL	• Prior ET • BM fibrosis 2-3
Minor Criteria			
• Sub-normal EPO level	• Anemia / loss of need for cytoreduction • LE blood picture • Increasing splenomegaly • Constitutional symptoms	• Clonal marker or absence of reactive cause	• Anemia / drop in Hb • LE blood picture • Increasing splenomegaly • Constitutional symptoms • Elevated LDH



WHO and ICC Criteria 2022

Pre-PMF	Overt PMF
Major Criteria	
• Mega proliferation and atypia, fibrosis <2, increased BM cellularity, granulocytic proliferation, reduced erythropoiesis • Not meeting criteria for other MDS/MPN • JAK2, CALR or MPL mutations or other clonal markers or no reactive cause	• Mega proliferation and atypia, fibrosis >= 2 • Not meeting criteria for other MDS/MPN • JAK2, CALR or MPL mutations or other clonal markers or no reactive cause
Minor Criteria	
• Anemia • WCC >= 11 x10⁹/L • Splenomegaly • Raised LDH	• Anemia • WCC >= 11 x10⁹/L • Splenomegaly • Raised LDH • LE blood picture



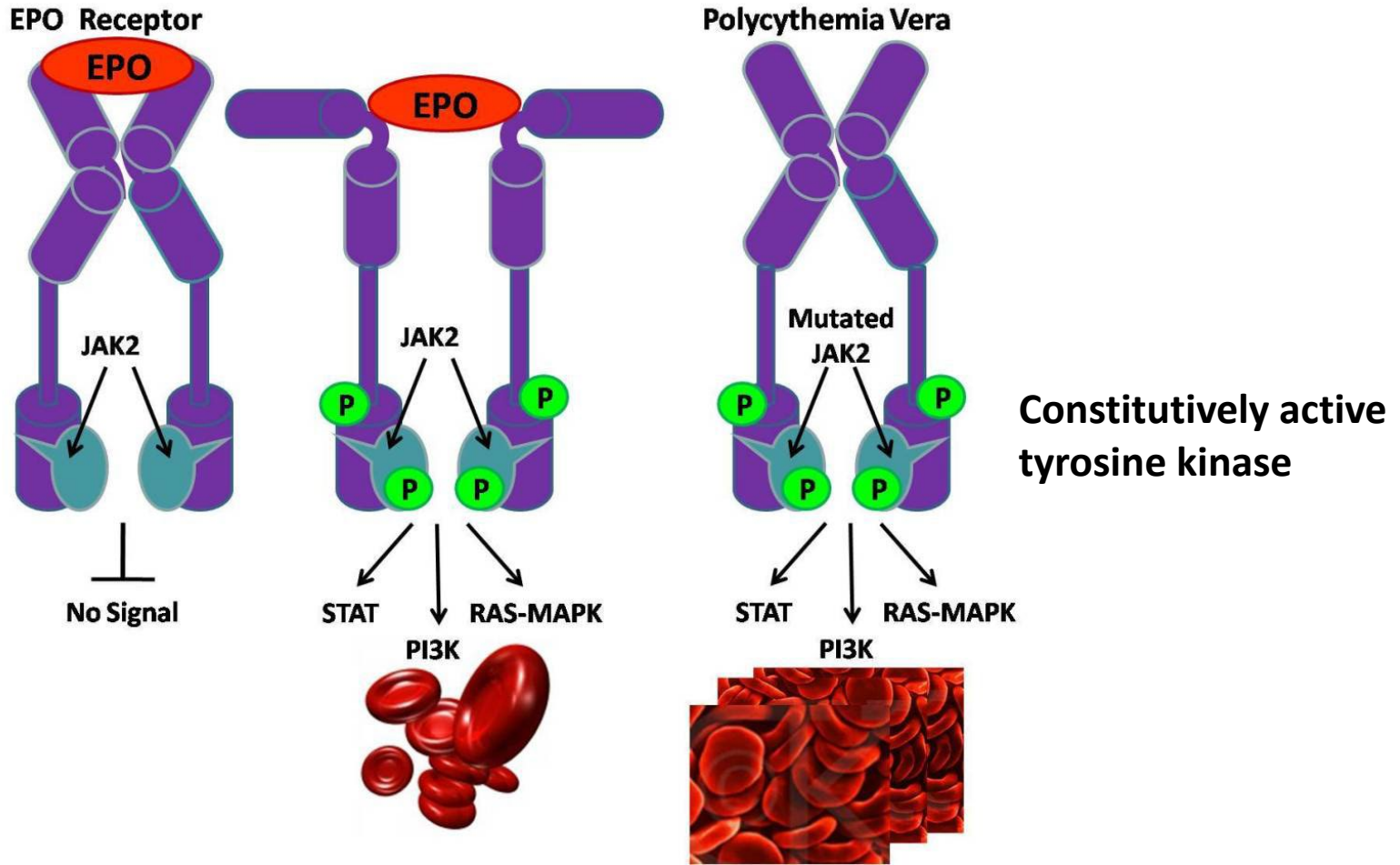
Management of classical Philadelphia chromosome-negative myeloproliferative neoplasms in Asia: consensus of the Asian Myeloid Working Group

Harinder Gill^{1,15} · Garret M. K. Leung¹ · Melissa G. M. Ooi^{2,3} · Winnie Z. Y. Teo^{2,4} · Chieh-Lee Wong⁵ · Chul Won Choi⁶ · Gee-Chuan Wong⁷ · Zhentang Lao⁷ · Ponlapat Rojnuckarin⁸ · Ma. Rosario Irene D. Castillo⁹ · Zhijian Xiao¹⁰ · Hsin-An Hou¹¹ · Ming-Chung Kuo¹² · Lee-Yung Shih¹² · Gin-Gin Gan¹³ · Chien-Chin Lin¹⁴ · Wee-Joo Chng^{2,3} · Yok-Lam Kwong¹

Received: 18 March 2023 / Accepted: 4 September 2023

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JAK2 mutation in MPNs



- Translocate to nucleus
- Activate transcription of target gene

POLYCYTHAEMIA VERA



Hb >16.5g/dL
Hct >49%



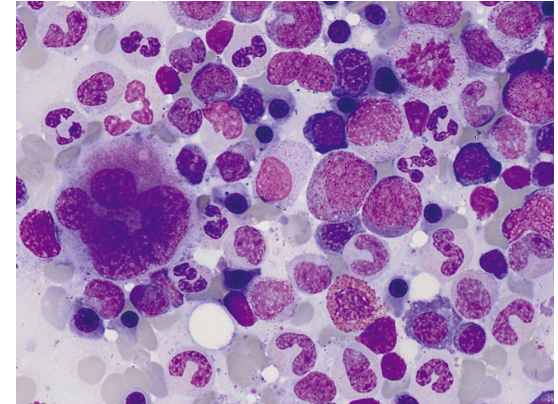
Hb >16g/dL
Hct >48%

Exclude secondary causes

1. **Absolute** polycythaemia (increased red cell mass)
 - a) Primary e.g. Polycythaemia vera
 - b) Secondary (**EPO** mediated)
 - i. Hypoxia e.g. COPD, cyanotic heart disease
 - ii. Abnormal EPO secretion, e.g. hepatoma
2. **Apparent** polycythaemia (normal red cell mass)
 - Normal red cell mass/low plasma volume
 - => elevated HCT

Diagnosis by BM and JAK2 mutation (95%)

Low EPO



Clinical features

Hyper-viscosity symptoms: dizziness, headaches

Pruritus particularly after a hot bath

Risk of arterial and venous thrombosis
Microvascular symptoms

Treatment

Venesection

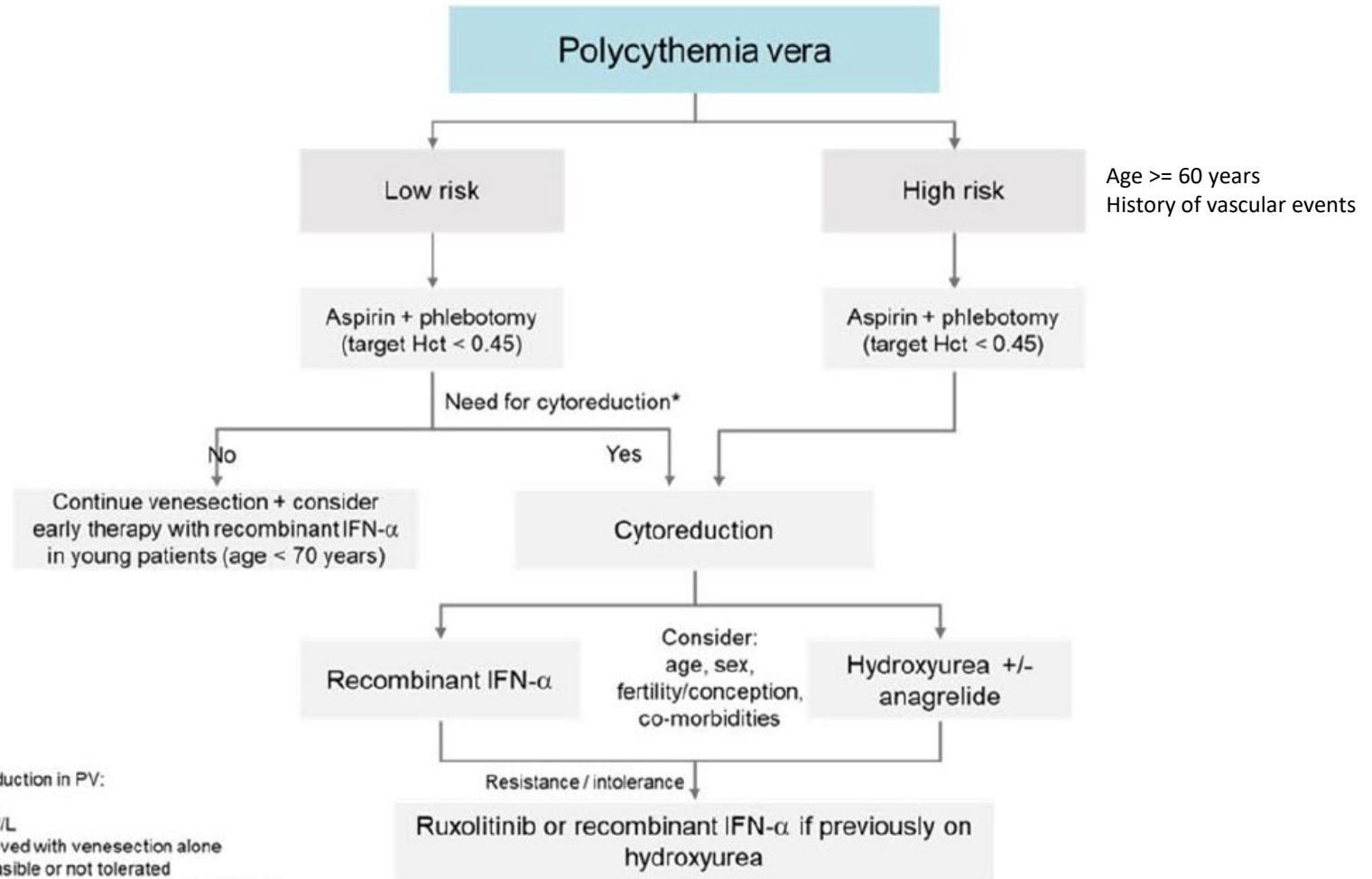
Cytoreductive: hydroxyurea, Interferon

Aspirin

***Transformation to myelofibrosis (10%), acute leukaemia (5%)**

Treatment Algorithm

A



*Indications for cytoreduction in PV:

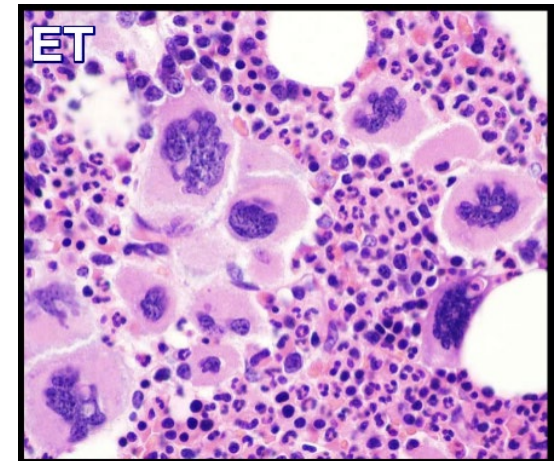
- WBC $> 15 \times 10^9/L$
- Platelet $> 450 \times 10^9/L$
- Hct target not achieved with venesection alone
- Venesection not feasible or not tolerated
- MPN-SAF TSS ≥ 20 or pruritus score ≥ 5 on MPN-SAF
- Vasomotor symptoms

ESSENTIAL THROMBOCYTHAEMIA ➡ Plt persistently >450

1. Usually asymptomatic
2. **Microvascular symptoms:** headache, dizziness, erythromelalgia
3. **Thrombosis:** Arterial > Venous

Diagnosis by exclusion

- => infections, inflammations, malignancy, iron deficiency
- ⇒ Bone marrow to exclude CML and other MPN/MDS
- ⇒ JAK2 V617F mutation in ~50%, CALR, MPL



Management => thrombosis risk

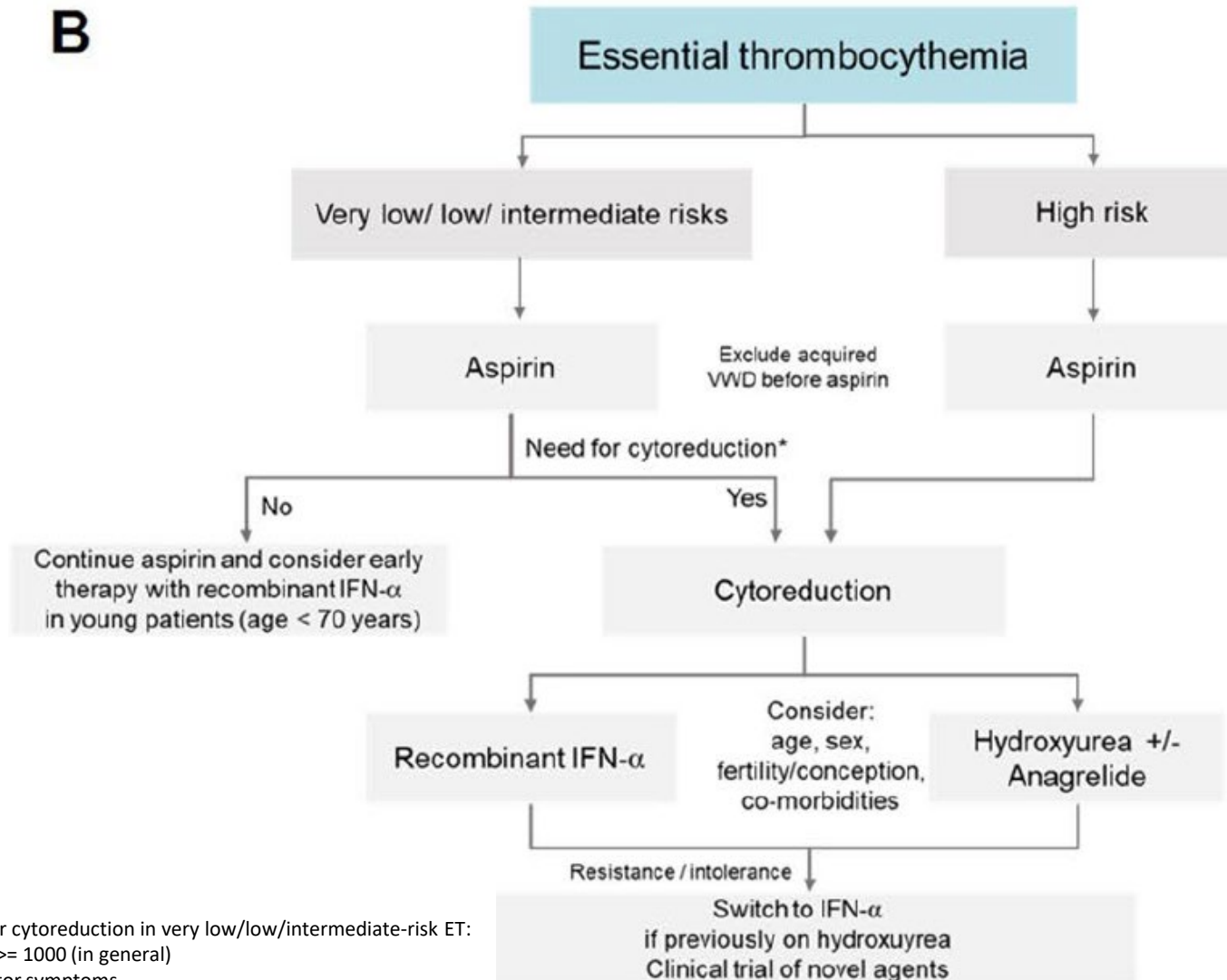
1. Age > 60 years	HU and aspirin, IFN (pregnancy) Cardiovascular risk modification (HT, smoking, CHF, WBC)
2. History of thromboembolic disease	

***Paradoxical bleeding in very high platelet (acquired VWD)**

****Risk of transformation to myelofibrosis and AML (<5%)**

Treatment Algorithm

B



*Indications for cytoreduction in very low/low/intermediate-risk ET:

- Platelet ≥ 1000 (in general)
- Vasomotor symptoms

MYELOFIBROSIS

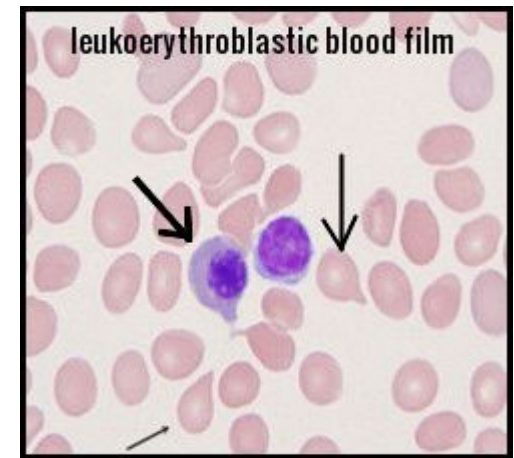
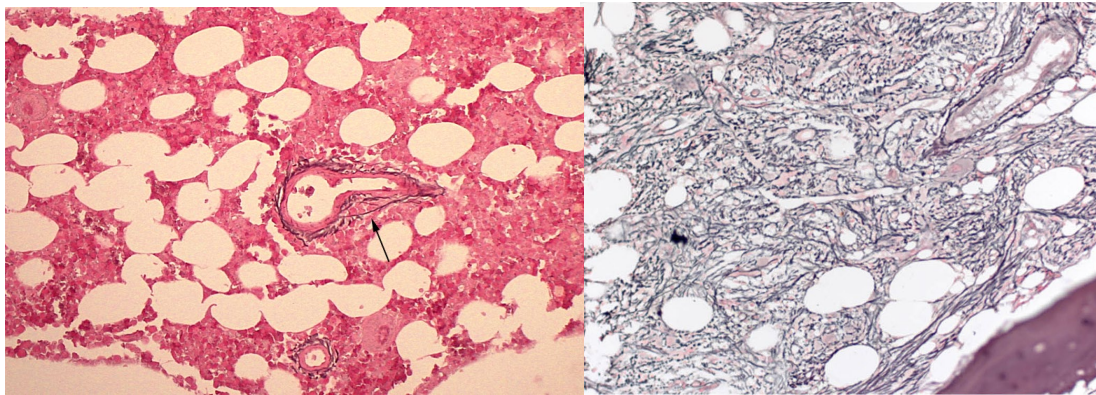
- MPN characterized by bone marrow fibrosis and **splenomegaly**
- **PMF fibroblasts are not clonal**
 - Abnormal megakaryocytes release cytokines (PDGF and TGFb) which stimulate fibroblast proliferation and collagen deposition
 - Fibrotic BM unable to function and HSC move to spleen and liver (**extramedullary haematopoiesis**)

No significant or functional haematopoiesis occurs even in this situation, and removal of the spleen will not impair the blood production capability.

- Patients presented with fatigue, weight loss (hypercatabolic), massive splenomegaly, portal hypertension

MYELOFIBROSIS

- Pancytopenia with **leukoerythroblastic** blood picture
- Bone marrow dry tap, trephine showing dense reticulin

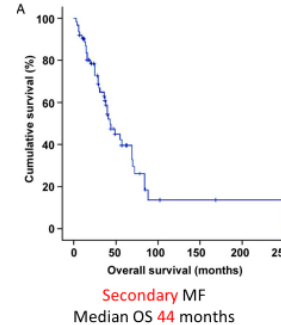
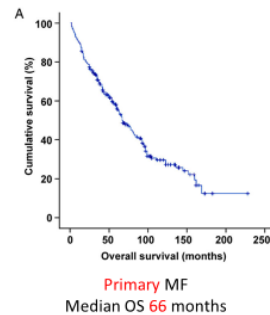


- JAK2 mutation 50%, other mutations

- **Leukaemic transformation in 15%**
- **Median survival 3-5 years**
- **Allogeneic stem cell transplantation** is the only potentially curative procedure - feasibility limited by age and comorbidities

Epidemiology

Survival of Myelofibrosis in Hong Kong



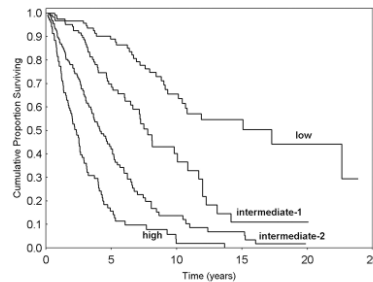
Gill H, et al., Hematology 2016.

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Risk Stratification Primary Myelofibrosis

DIPSS

- Age >65
- WCC >25 x10⁹/L
- Hb <10 g/dL
- PB blasts >= 1%
- Constitutional symptoms



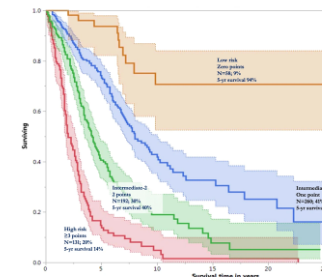
Passamonti F, et al., Blood 2010.

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Risk Stratification Primary Myelofibrosis

GIPSS

- Karyotype
- Absence of type1/like CALR
- ASXL1
- SRSF2
- U2AF1 Q157



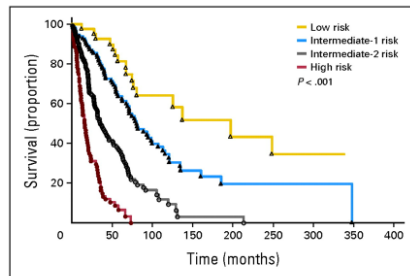
Tefferi a, et al., Leukemia 2018.

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Risk Stratification Primary Myelofibrosis

DIPSS-plus

- Age >65
- WCC >25 x10⁹/L
- Hb <10 g/dL
- PB blasts >= 1%
- Constitutional symptoms
- Unfavourable karyotype
- Platelet <100
- Transfusion dependency



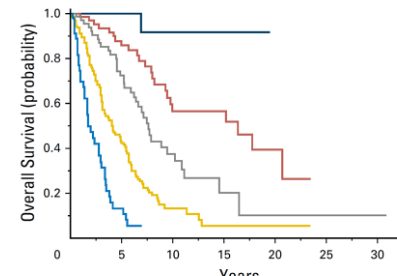
Gangat N, et al., J Clin Oncol 2011.

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Risk Stratification Primary Myelofibrosis

MIPSS70+ v2.0

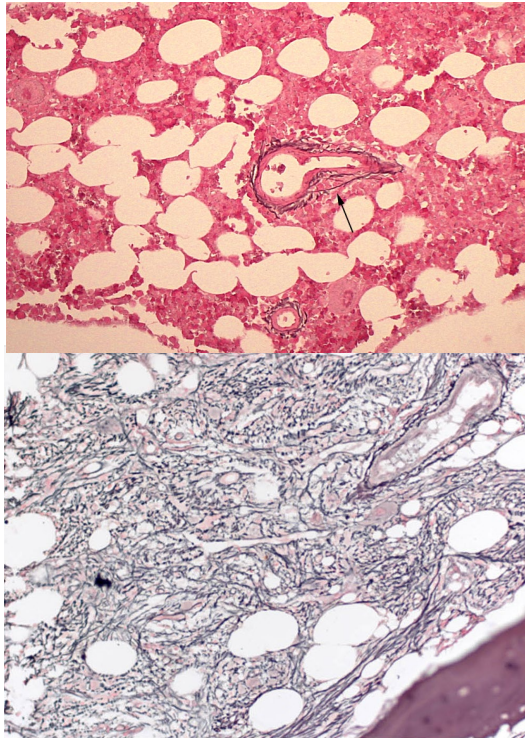
- Anemia severity
- PB blasts
- Constitutional symptoms
- Karyotype
- Absence of type 1/like CALR mutation
- HMR mutations



Tefferi A, et al., J Clin Oncol 2018

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MYELOFIBROSIS

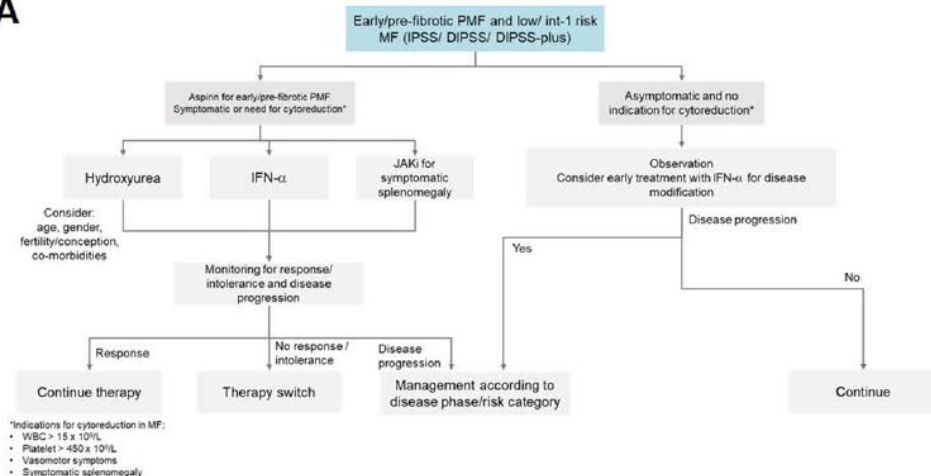


- **Supportive** for cytopenia
 - Transfusion (+/- iron chelation)
 - Epo
 - TGF-b inhibitor e.g. luspatercept
- **Ruxolitinib**, a JAK1/2 kinase inhibitor
 - decrease size of spleen and constitutional symptoms
- **Splenectomy** sometimes performed
 - to relieve pressure symptoms and decrease transfusion requirement

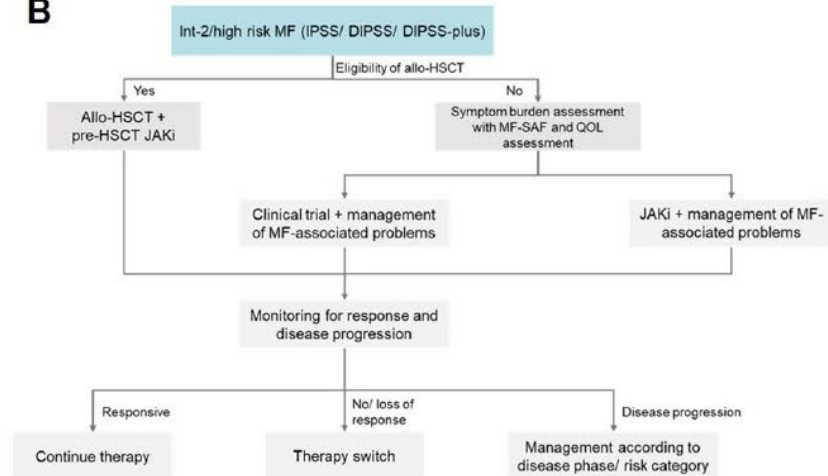
Jak inhibitor	Approved agents			Investigational agents		
	Ruxolitinib	Fedratinib	Pacritinib	Momelotinib	Ilginatinib	Itacitinib
Targets	Jak1, Jak2	Jak2, Jak1, FLT3, BRD4, TYK2, many others	Jak2, IRAK1, FLT3	Jak1, Jak2, ACVR	Jak2	Jak1
Indications	Symptomatic MF with platelets $\geq 50 \times 10^9/L$	Symptomatic MF with platelets $\geq 50 \times 10^9/L$	Symptomatic MF with platelets $< 50 \times 10^9/L$	In development for MF with anemia	In development for MF with thrombocytopenia	In development for MF with anemia

Treatment Algorithm

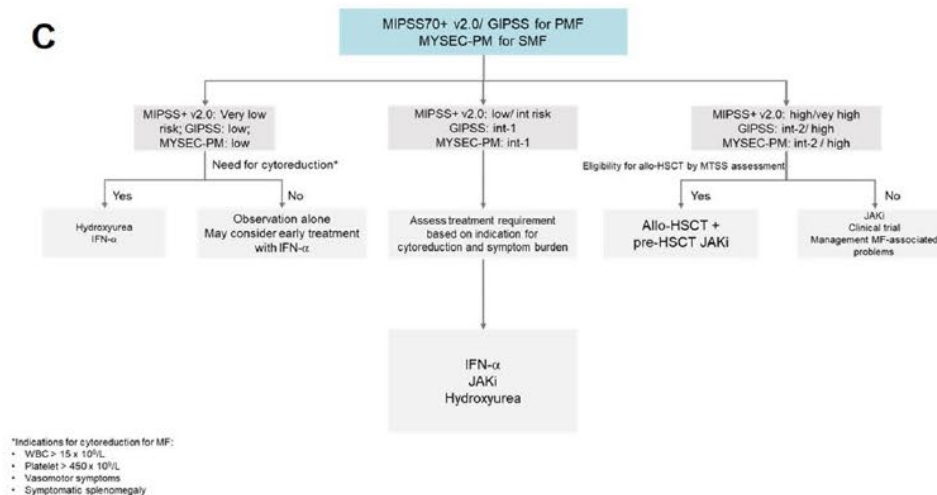
A



B



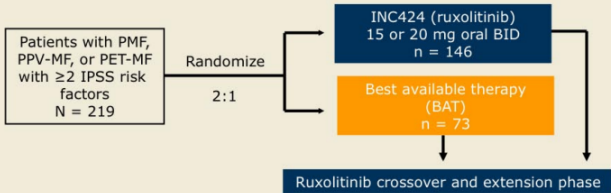
C



Ruxolitinib

COMFORT-II: Phase III Trial of JAK1/JAK2 Inhibitor Ruxolitinib versus BAT

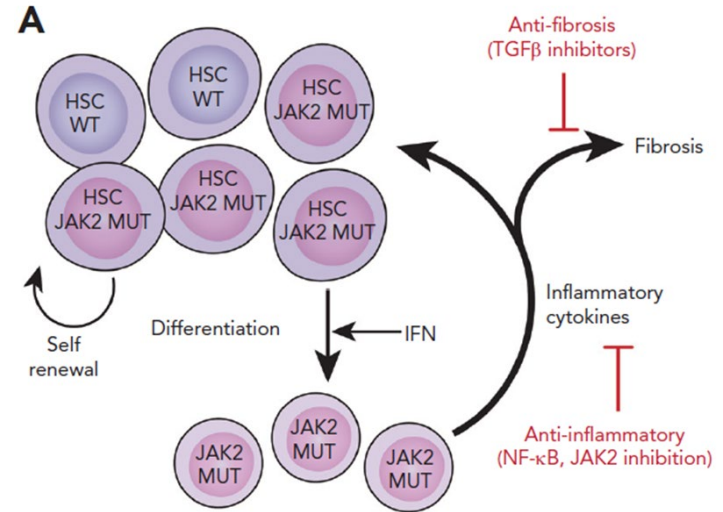
- Randomized, open-label, multicenter Phase III trial
- Patients were stratified based on baseline IPSS risk category



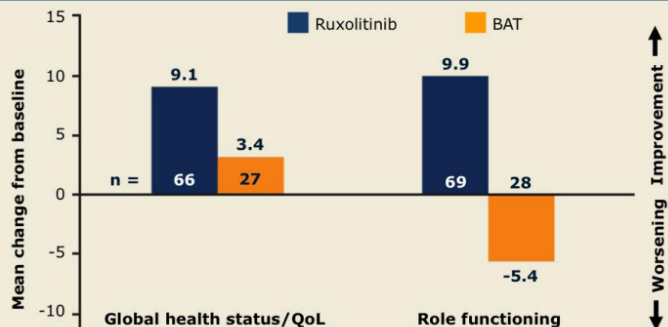
- Progression events that qualified for the crossover and extension phase:
 - Splenectomy
 - Progressive splenomegaly as defined by a 25% increase in spleen volume compared with the on-study nadir (including baseline)

Harrison CN et al. *Proc ASCO* 2011;Abstract LBA6501.

Research To Practice®



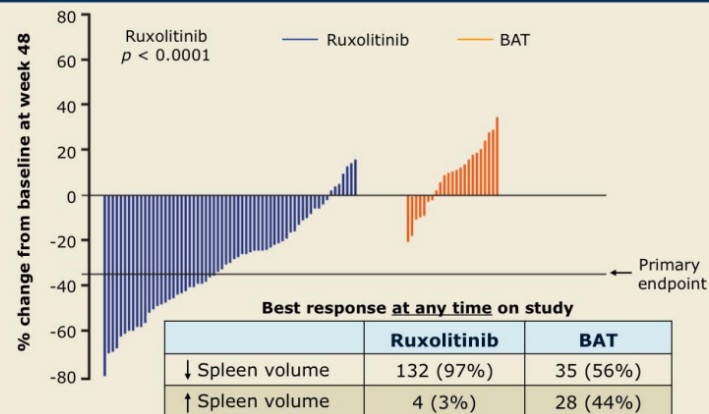
Change in EORTC QLQ-C30 Scores at Week 48 Compared to Baseline



• Improvements were seen by week eight and continued through week 48.

With permission from Harrison CN et al. *Proc ASCO* 2011;Abstract LBA6501.

Percent Change in Spleen Volume from Baseline

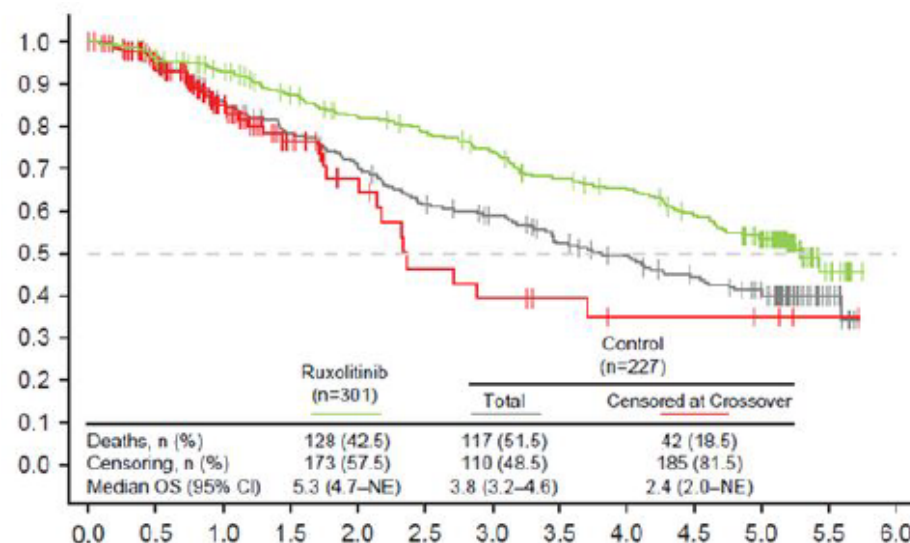
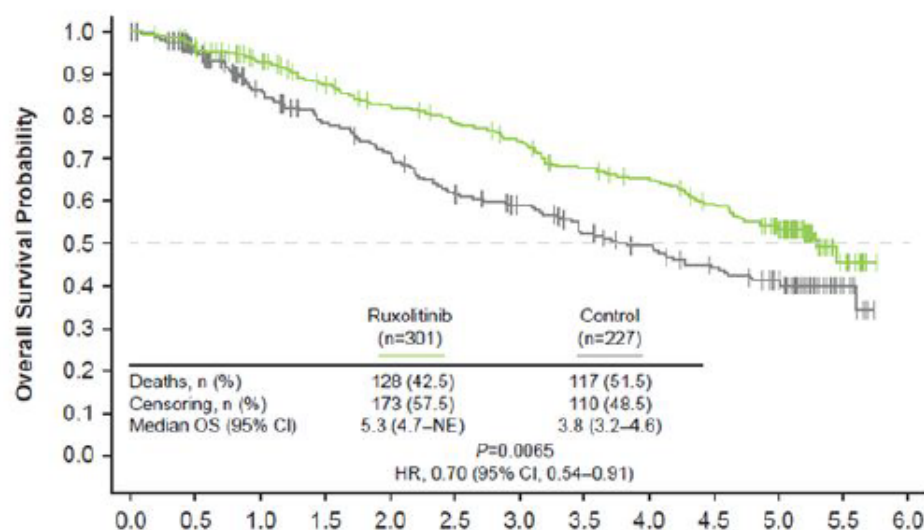


With permission from Harrison CN et al. *Proc ASCO* 2011;Abstract LBA6501.

<http://www.researchtopractice.com/5MJCASCO2011/Myelofibrosis/2>

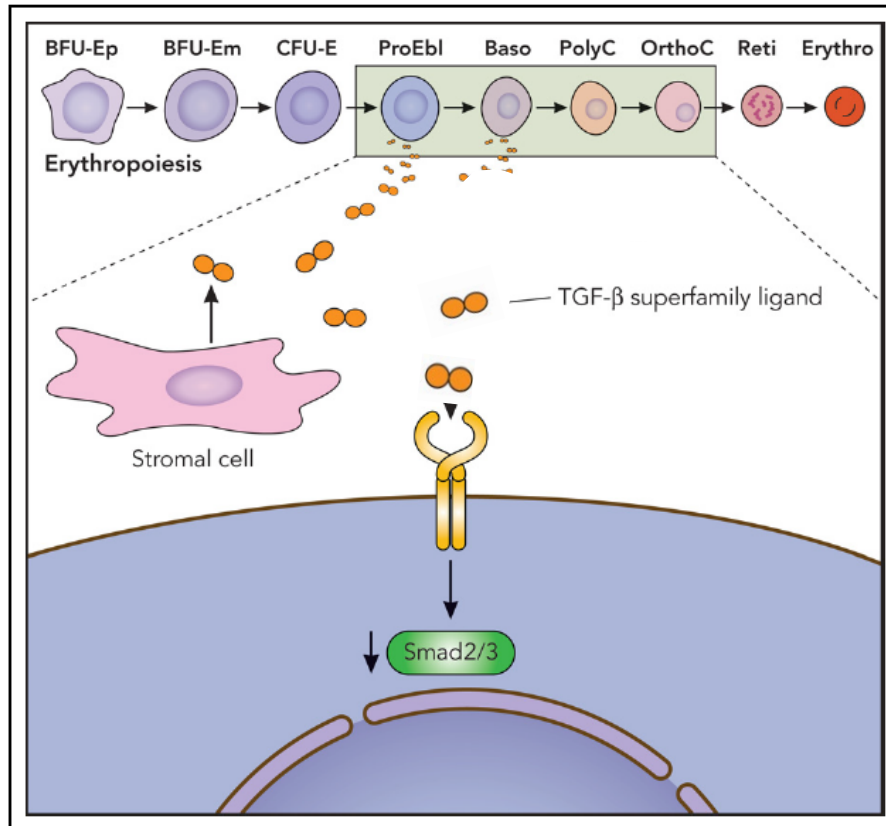
Joan How et al. *Biology and therapeutic targeting of molecular mechanisms in MPNs. Blood* 2023

A Pooled Overall Survival (OS) Analysis of 5-Year Data from the COMFORT-I and COMFORT-II Trials of Ruxolitinib for the Treatment of Myelofibrosis (MF)

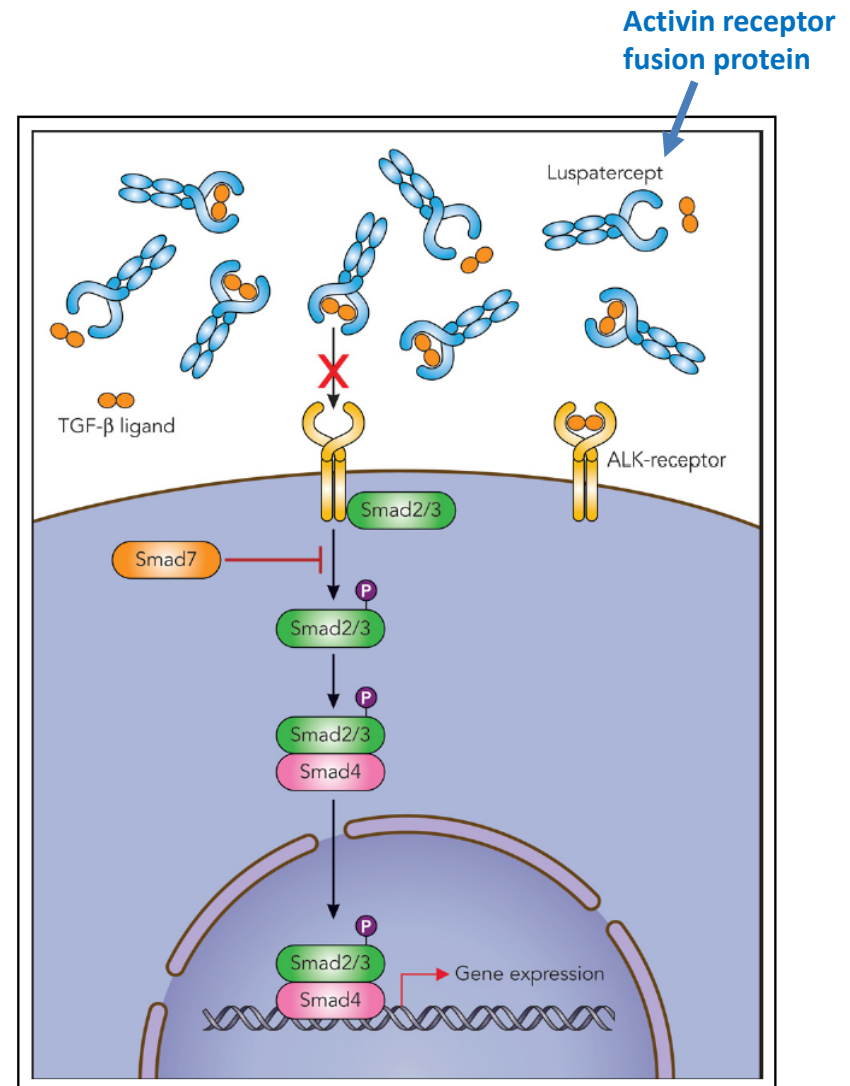


Luspatercept is a specific activin receptor fusion protein that acts as a ligand trap to neutralize negative regulators of late-stage erythropoiesis.

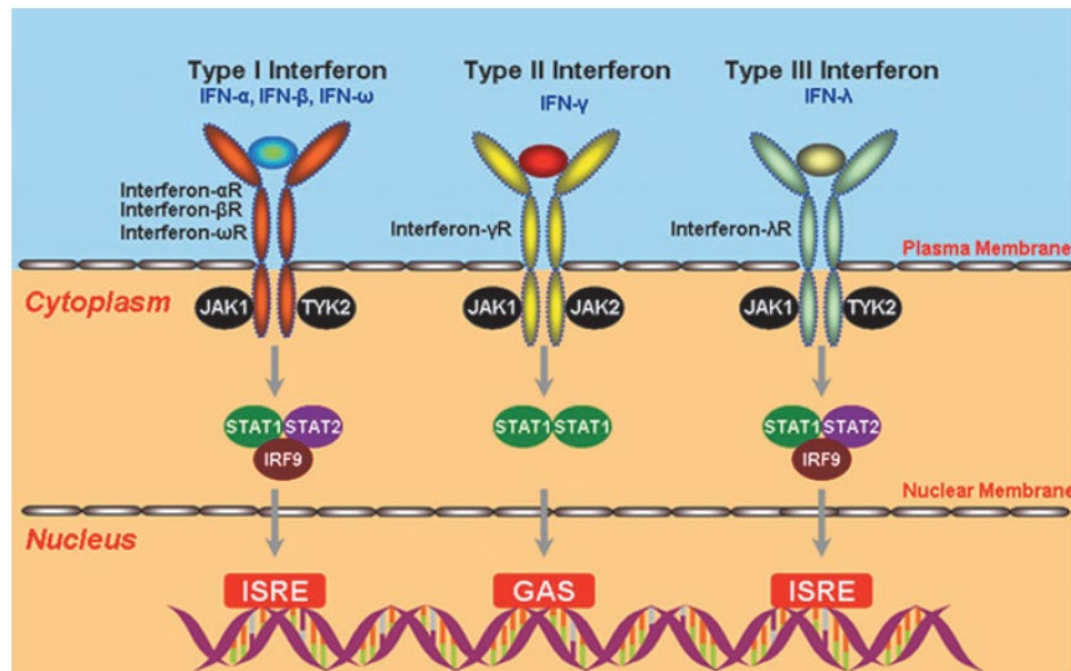
In the TGF- β superfamily, SMADs are important physiologic regulators of hematopoiesis.



1. After ligand binding and receptor phosphorylation, the SMAD2/3 signaling pathway becomes activated and mediates intracellular signaling;
2. E.g. In MDS, SMAD2/3 downstream mediators are constitutively activated and overexpressed in MDS CD34 cells, which leads to impaired terminal erythroid differentiation and subsequent anemia as a result of an inhibitory effect on red-cell maturation.



How does Interferon work in the Classical BCR-ABL-Negative MPNs



IFNs has **ANT-ANGIOGENIC, ANTI-PROLIFERATIVE, APRO-APOPTOTIC, and IMMUNOMODULATORY** effects.

Mechanisms of Action of IFN

- Inhibition of proliferation**
 - IFN-α and IFN-β can interfere with all mitotic cell cycle phases, primarily within G1
- Pro-apoptosis**
 - Inducing TNF-related apoptosis inducing ligand (TRAIL) and/or Fas/FasL mediated apoptosis
- Immunomodulatory properties**
 - Promote the activity of T lymphocytes, monocytes, NK cells, and macrophages
 - augment the expression of MHC antigens and tumor associated antigen
- Alteration of gene expression (IFN-inducible genes)**
- Inhibition of hematopoiesis**
 - IFN-α inhibit normal BM haematopoietic progenitors
 - Inhibit normal cytokine production e.g. GM-CSF

Rationale for Using IFN in MPNs

- Suppression of Megakaryopoiesis**
 - Suppress colony-forming unit–megakaryocyte (CFU-Mk)
 - IFN-α can inhibit the secretion of thrombopoietin (TPO), and also inhibit TPO-induced phosphorylation of JAK2 and STAT5
- Inhibition of proliferation of human Erythroid colonizing forming cells (ECFCs)**
 - IFNs, in particular IFN-γ, can inhibit growth and differentiation of ECFCs and stimulate apoptosis in erythroblasts
 - By its **regulatory effects on T cells and monocytes**, IFNs can mediate secretion of multiple **lymphokines and cytokines** that have inhibitory effects on erythroid colony cells
- Decrease in the human bone marrow Fibroblasts**
 - IFN-γ through its effects on the STAT pathway can modulate the production of this **cytokines**, resulting in less formation of collagen and thereby less fibrotic changes in the bone marrow, which can be seen in the MPN, particularly MF

Table 1 AMWG recommendations for the initial assessment and investigations in Ph-negative myeloproliferative neoplasms

History and physical examination
Assessment of prior or current thrombotic and hemorrhagic events
Assessment of cardiovascular risk factors
Medical history, psychiatric history, transfusion history
Exclusion of reactive causes of thrombocytosis or leucocytosis
Assessment of hepatosplenomegaly
CBC with differential count and blood film review
Serum electrolytes, RFT, LFT, LDH, urate
Serum EPO level for polycythemia
Iron profile
Clotting profile in patients with thrombocytosis
vWF assays (for the exclusion of acquired vWD in patients with thrombocytosis or bleeding tendency)
Hepatitis B serology including HBsAg/Ab and anti-HBc IgG
HBV DNA in patients who are HBsAg or anti-HBc IgG positive
Hepatitis C serology
Thyroid function tests in candidates for IFN- α
Autoimmune markers (e.g. ANA and RF) and inflammatory markers (e.g. ESR and CRP) in the appropriate
CXR
Exclusion of active or latent tuberculosis in candidates for ruxolitinib
Other investigations to exclude reactive causes guided by medical history and physical examination
Bone marrow aspirate and trephine biopsy for morphologic assessment, reticulin staining and trichrome staining
Karyotype
<i>JAK2V617F</i> assessment
<i>JAK2</i> exon 12 assessment in PV negative for <i>JAK2V617F</i>
<i>CALR</i> and <i>MPL</i> assessment in ET and PMF negative for <i>JAK2V617F</i>
RT-PCR or FISH for <i>BCR::ABL1</i> to exclude CML in MPN negative for <i>JAK2/CALR/MPL</i> mutations
NGS for myeloid gene panel
For risk stratification in all patients with PMF or SMF who are potential candidates for allo-HSCT
To establish clonality in patients with MPN who are negative for <i>JAK2/CALR/MPL</i> mutations
Assessment of symptom burden using MPN-SAF
HLA typing in patients with PMF and SMF who are potential candidates for allo-HSCT

Initial workup for suspected MPN

CBC: complete blood count; RFT: renal function test; LFT: liver function test; LDH: lactate dehydrogenase; EPO: erythropoietin; vWF: von Willebrand Factor; vWD: von Willebrand disease; HBsAg/Ab: hepatitis B surface antigen/antibody; anti-HBc IgG: anti-hepatitis B core IgG; ANA: anti nuclear antibodies; RF: rheumatoid factor; ESR: erythrocyte sedimentation rate; CRP: C reactive protein; CXR: chest X-ray; DV:

Take home message

Splenomegaly / MPN

- **Many** causes of splenomegaly
- **Size** of the spleen and how common the diseases can help narrow down differential diagnosis
- **Geographical** location determines the possible infective causes of splenomegaly
- **POST SPLENECTOMY INFECTIONS** requires urgent treatment
- Brief review of the MPNs: PV, ET, PMF



THANK YOU

QUESTIONS ?

https://eform.med.hku.hk/redirectSvr/tl/MBBSEvaluationForm?blockCode=JC_2324

TELL US WHAT
JC Whole Class Session
YOU THINK

THANK YOU!



Look for today's teacher
and the others you may have missed.
(HKU Portal login required)